W125–W134 Calibration Report: Per-Site Flushing Worsens RMSE—and Why

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March 8, 2026

1 Executive Summary

Per-site flushing derived from enclosedness metrics **worsened calibration fit** across all configurations tested. The best run (W129, $\alpha_{\text{env}} = 0.25$) achieved RMSE=0.691, compared to W117's 0.603 under uniform $\phi = 0.5$. This report argues this is not a bug—it reveals a **fundamental asymmetry in how flushing affects disease dynamics**, and points toward the missing mechanism (salinity).

2 The Argument

2.1 What Per-Site Flushing Changed

Under uniform $\phi = 0.5$, all 896 sites flushed pathogen at the same rate. The north-south recovery gradient was driven entirely by SST-modulated α_{env} (warm south → more disease amplification).

Per-site flushing introduced two changes simultaneously:

1. **Enclosed sites (AK fjords, Salish Sea)**: ϕ dropped from 0.50 → 0.28–0.30
2. **Open coast (CA, OR)**: ϕ rose from 0.50 → 0.66–0.68

2.2 Why This Made Things Worse

The P_{env} dynamics at each node follow:

$$\frac{dP}{dt} = \underbrace{\text{shedding}}_{\text{infected hosts}} - \underbrace{\xi \cdot P}_{\text{decay}} - \underbrace{\phi \cdot P}_{\textbf{flushing}} + \underbrace{\text{dispersal}}_{\text{D matrix}} + \underbrace{\text{floor}}_{\alpha_{\text{env}}} \tag{1}$$

Lower ϕ means pathogen decays more slowly locally. For enclosed AK fjord sites:

- ϕ dropped from 0.5 to ∼0.3 → local P_{env} accumulates 67% more
- Disease pressure *increases* → populations recover *less*
- AK-PWS recovery: 7.85% (W117) → 3.74% (W129)—**moving away from the 50% target**

For open coast CA/OR sites:

- ϕ rose from 0.5 to ∼0.67 → pathogen flushes 34% faster

- Disease pressure *decreases* → populations recover *more*
- CA-N recovery: 0.19% (W117) → 0.43% (W129)—**moving away from the 0.1% target**

Both effects push RMSE upward. Per-site flushing counteracts the SST-driven gradient: it makes the north sicker (where we need recovery) and the south healthier (where we need suppression).

2.3 The D Matrix Partially Compensates

The pathogen dispersal matrix is modulated by source ϕ:

$$D[j,k] = \phi_j \cdot f_{\text{out}} \cdot \exp(-d_{jk}/D_P) \cdot S_{jk}$$

So enclosed sites (low ϕ_j) export less pathogen. This *should* protect enclosed neighbors. But the effect is asymmetric:

- **Export reduction**: enclosed sites shed less to neighbors (helpful)
- **Local retention**: enclosed sites retain own P_{env} longer (**harmful, and dominant**)
- **Import unchanged**: incoming pathogen from the wavefront isn't modulated by recipient ϕ

Once disease arrives via the wavefront (which uses its own long-range kernel), local retention dominates. The wavefront D matrix is independent of ϕ—so the initial disease import is identical, but the local response differs dramatically.

2.4 The Real World is Different

Empirically, enclosed waterways like Prince William Sound *are* refuges (Gehman et al. 2025, Proc Royal Soc B). But the protection mechanism isn't flushing dynamics—it's:

1. **Salinity**: Freshwater lens in fjords (glacial melt, river input) directly kills *Vibrio* (halophilic; need >0.5% NaCl). PWS surface salinity drops to 20 PSU seasonally.
2. **Depth refuge**: Low salinity at surface pushes *Pycnopodia* deeper into cold water, below the *Vibrio* activity threshold.

Our model captures temperature (SST) but **not salinity**. Without salinity, per-site flushing can only encode one relationship: enclosed → retains pathogen → more disease. This is physically correct but biologically incomplete. The real protective mechanism (salinity killing *Vibrio*) is absent.

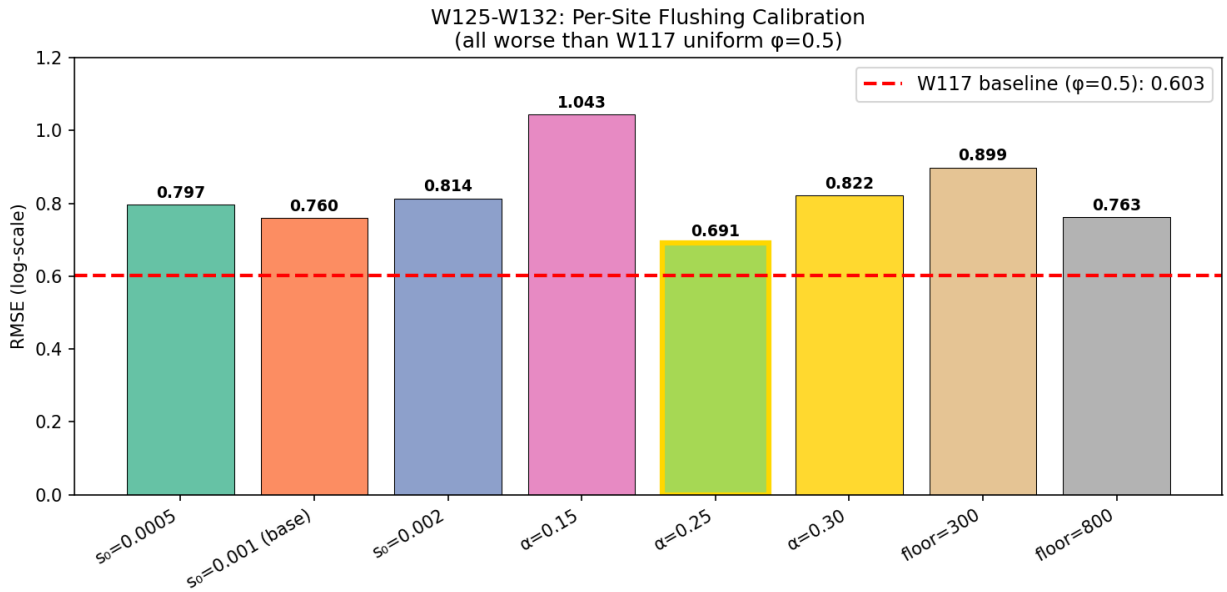

Figure 1: All W125–W132 runs score worse than the W117 baseline (uniform $\phi = 0.5$, dashed red line). W129 ($\alpha = 0.25$) is the best at 0.691 but still 15% worse than W117 (0.603).

3 Results

3.1 RMSE Comparison

3.2 Regional Breakdown

3.3 Recovery Trajectories

3.4 All Runs Heatmap

3.5 Per-Region Error Diagnosis

3.6 Alpha Sensitivity

4 Implications

4.1 Per-Site Flushing is Physically Correct but Incomplete

The enclosedness-derived ϕ values are defensible: enclosed waterways do retain water and pathogens longer. The problem is that our model lacks the **compensating mechanism** that makes enclosed waterways refuges in reality: *salinity inhibition of Vibrio*.

4.2 What n_connectivity Might Do

The W135–W144 round tests $n_{\text{connectivity}} = 0.5$ and 2.0. With $n = 0.5$, the ϕ values compress toward midrange (less extreme differentiation). This might soften the damage but won't fix the fundamental direction. With $n = 2.0$, the contrast sharpens—likely worse.

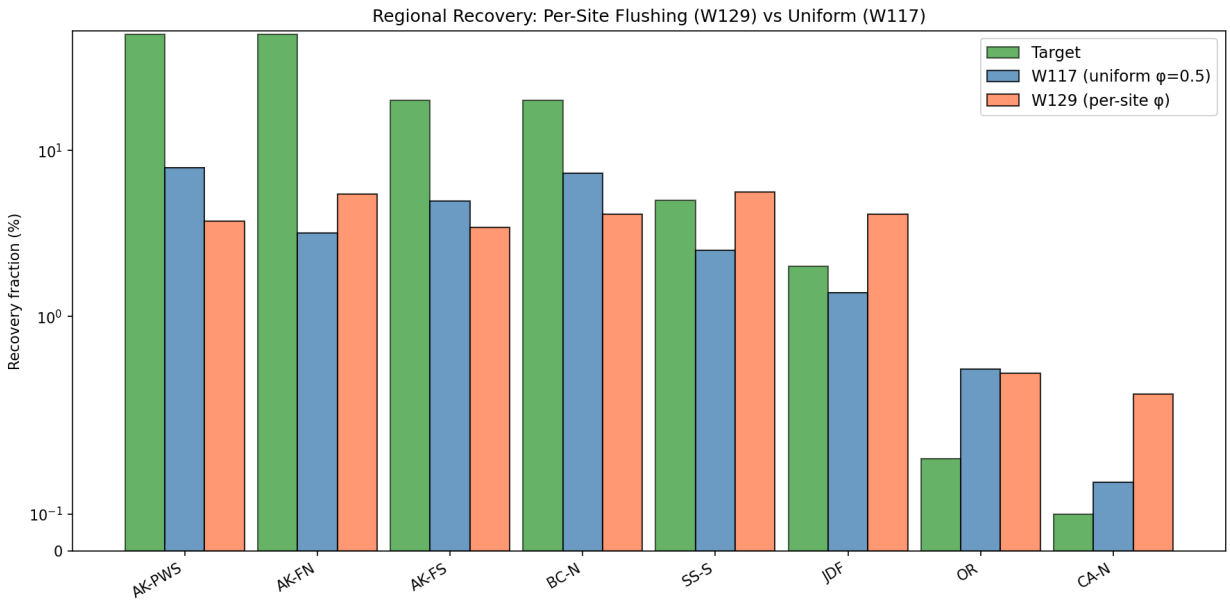

Figure 2: Per-region recovery comparison. W117 (blue) nails the south (SS-S, JDF, OR, CA-N all close to targets). W129 (orange) overshoots the south while undershoting the north—the gradient is flatter.

4.3 Possible Paths Forward

1. **Add salinity mechanism**: Per-node salinity modulation of *Vibrio* activity. Freshwater-influenced sites get reduced P_{env} amplification. This is the biologically correct fix (Gehman 2025).

2. **Decouple import vs retention**: Let ϕ modulate the D matrix (import/export) but use a uniform rate for local P_{env} decay. Enclosed sites would be protected from import without accumulating local pathogen.

3. **Reverse the sign**: Use enclosedness to *reduce* disease pressure rather than increase it (encode the net protective effect directly).

4. **Revert to uniform ϕ**: Accept that without salinity data, spatially-variable flushing hurts more than it helps.

Option 1 is the most scientifically defensible. Option 2 is a quick mechanical fix. Option 4 is the pragmatic choice if we want to move to SA and paper writing.

5 Summary Table

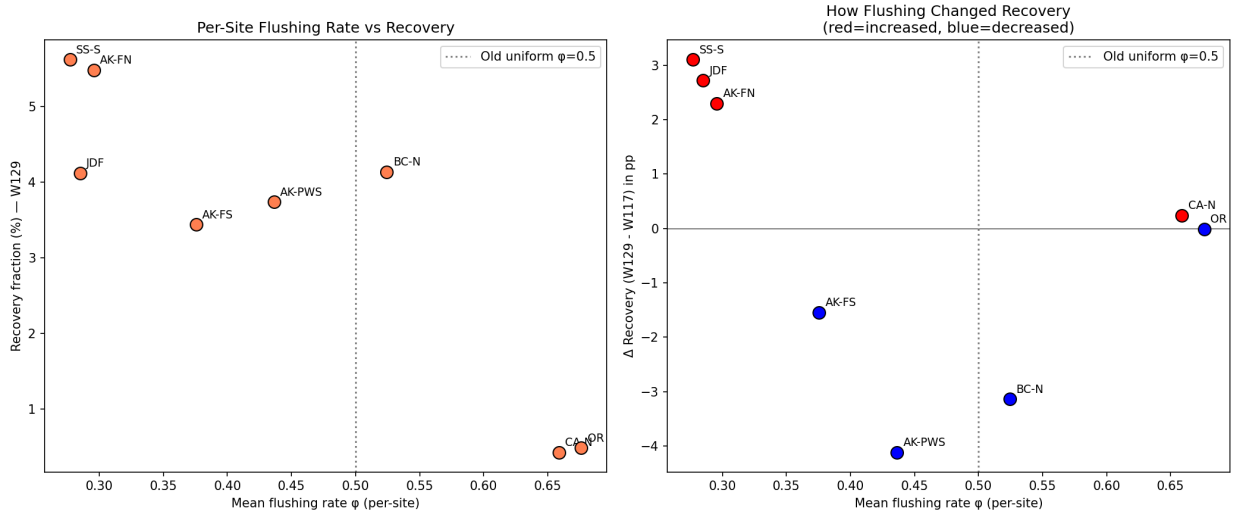

Figure 3: **Left**: Mean regional ϕ vs recovery. Low-ϕ regions (enclosed) have low recovery. **Right**: Change from W117. Regions that got lower ϕ (below old 0.5) lost recovery; regions that got higher ϕ gained recovery. The flushing change systematically pushes the gradient in the wrong direction.

Run	Config	RMSE	AK-PWS	AK-FN	BC-N	SS-S	CA-N
W117	uniform ϕ=0.5	**0.603**	7.85	3.19	7.27	2.52	0.19
W125	s_0=0.0005	0.797	7.95	12.59	8.12	11.03	2.17
W126	s_0=0.001	0.760	9.66	13.25	6.61	11.14	2.03
W127	s_0=0.002	0.814	8.97	11.00	7.98	8.81	2.88
W128	α=0.15	1.043	16.25	25.49	15.69	24.27	15.11
W129	α=0.25	0.691	3.74	5.48	4.14	5.62	0.43
W130	α=0.30	0.822	2.11	2.81	3.08	3.17	0.38
W131	floor=300	0.899	11.06	13.95	7.92	13.19	7.30
W132	floor=800	0.763	8.07	11.80	7.50	8.85	2.67
Target		—	50.0	50.0	20.0	5.0	0.1

Table 1: Recovery percentages at year 13 (seed-averaged). All per-site flushing runs are worse than W117.

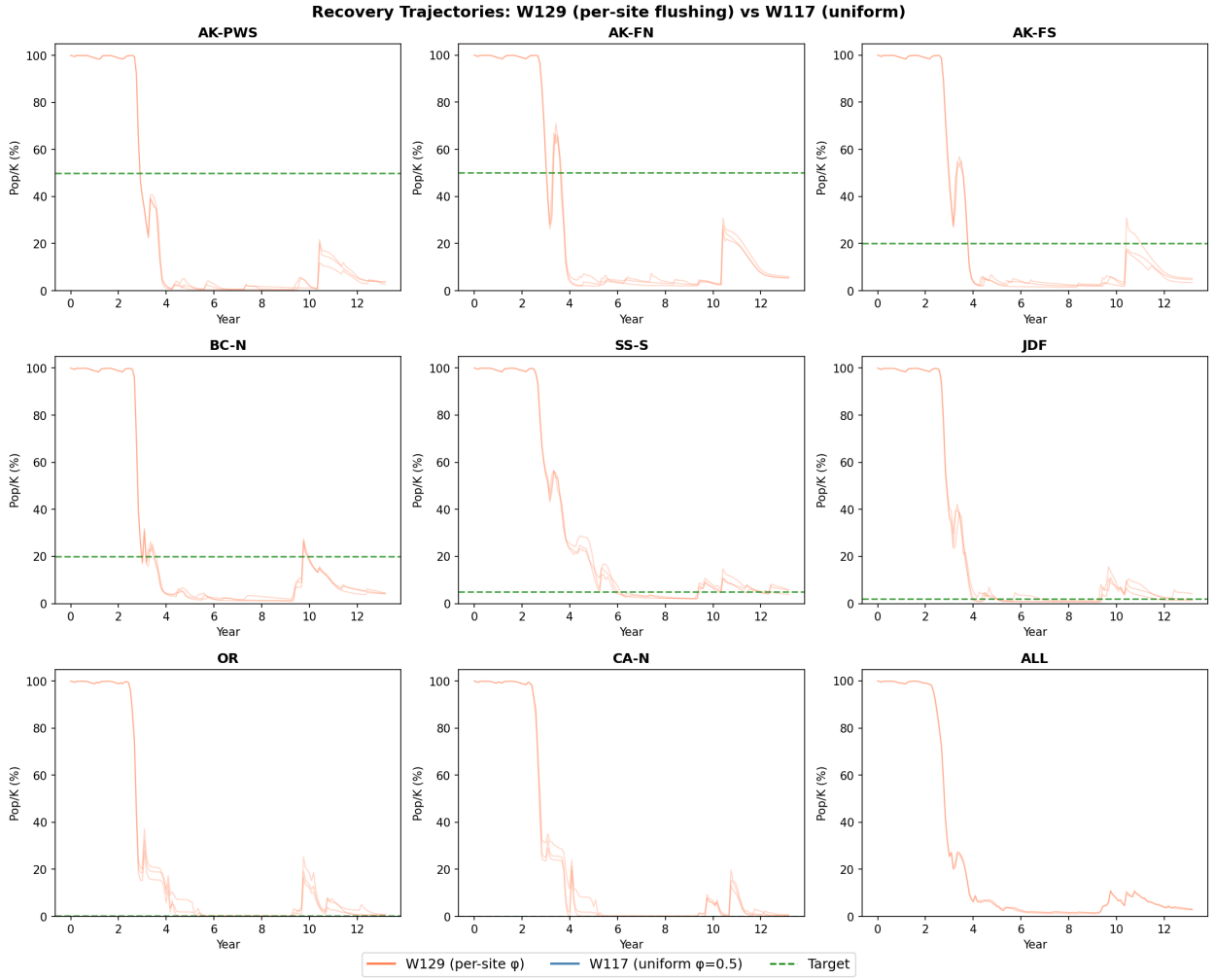

Figure 4: Monthly recovery trajectories for all 8 scored regions plus overall. Orange = W129 (per-site ϕ), blue = W117 (uniform). Dashed green = target. AK regions are worse under per-site flushing; southern regions show slightly more recovery.

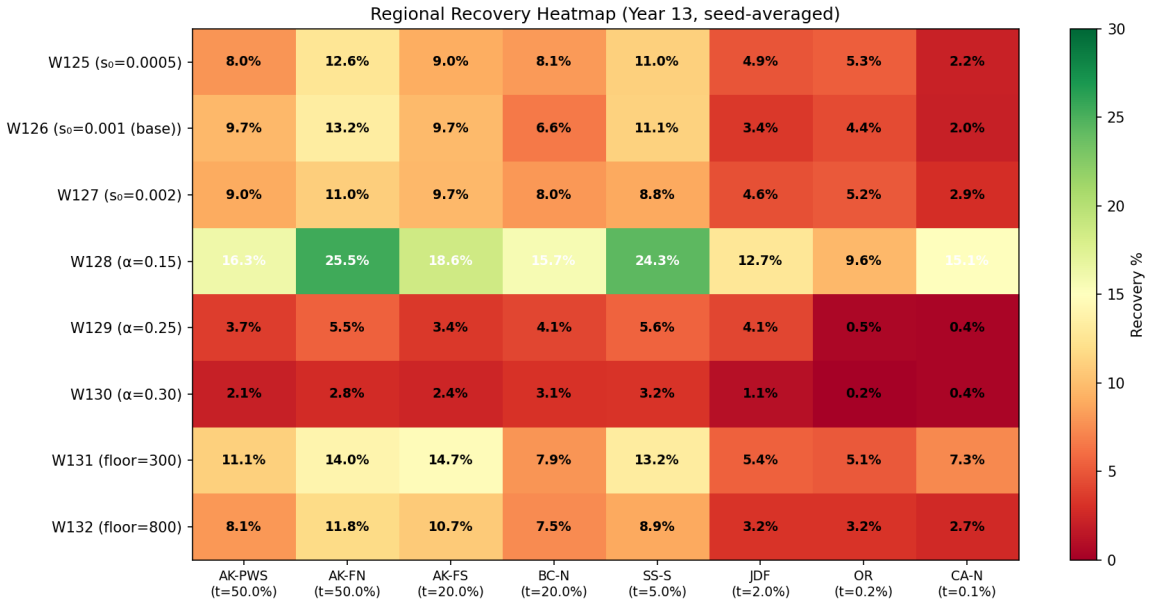

Figure 5: Regional recovery heatmap across all 8 runs. Note W128 ($\alpha = 0.15$, low disease amplification) has the highest AK recovery but also unacceptable southern recovery (>15% in CA-N). The gradient-forming mechanisms conflict.

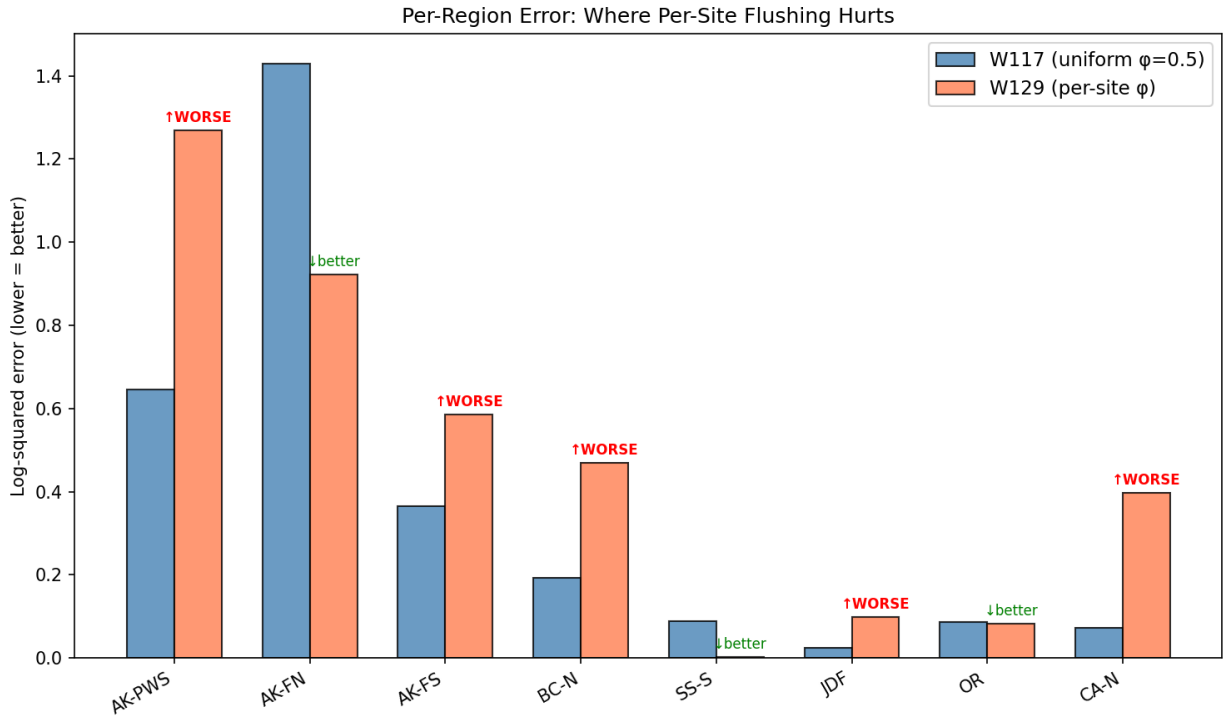

Figure 6: Where RMSE gets worse. W129 (orange) has higher error than W117 (blue) in AK-PWS, AK-FS, JDF, and CA-N. Only AK-FN and SS-S are comparable. The damage is spread across both ends of the gradient.

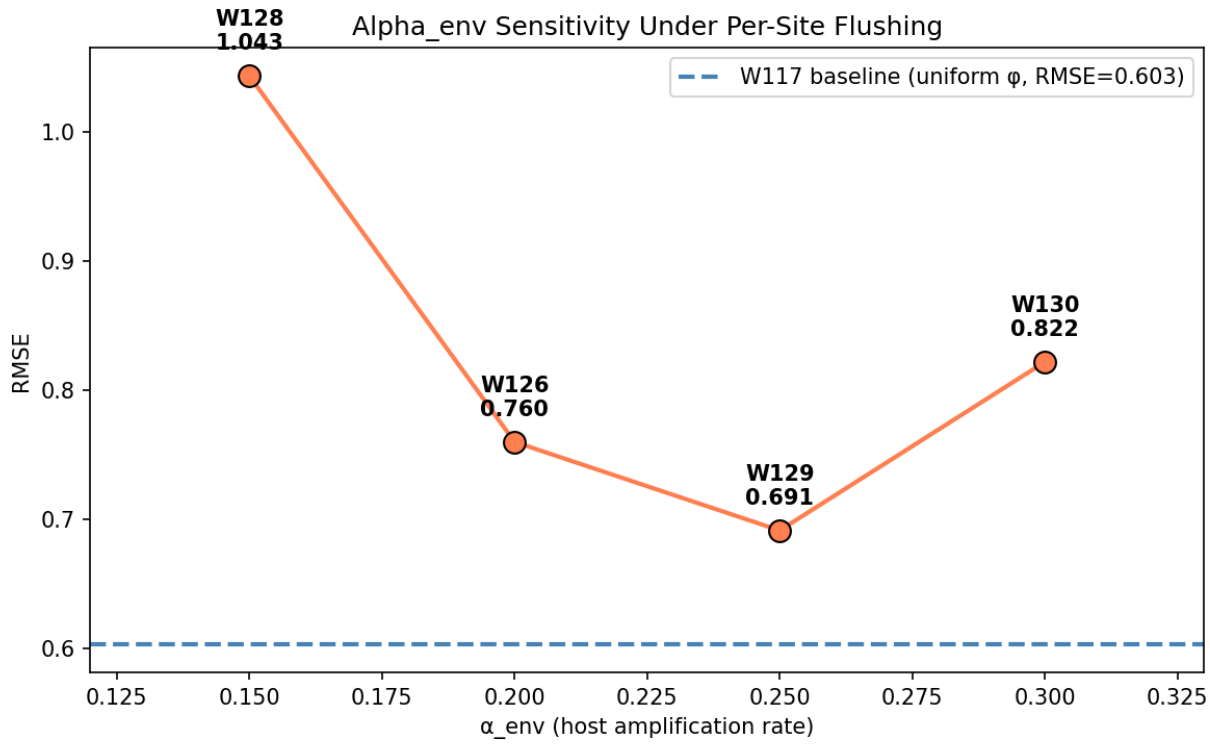

Figure 7: α_{env} sensitivity under per-site flushing. Optimal at 0.25 (W129), but no value reaches the W117 baseline. The U-shape suggests α trades off north vs south: too low lets south recover, too high suppresses everything.